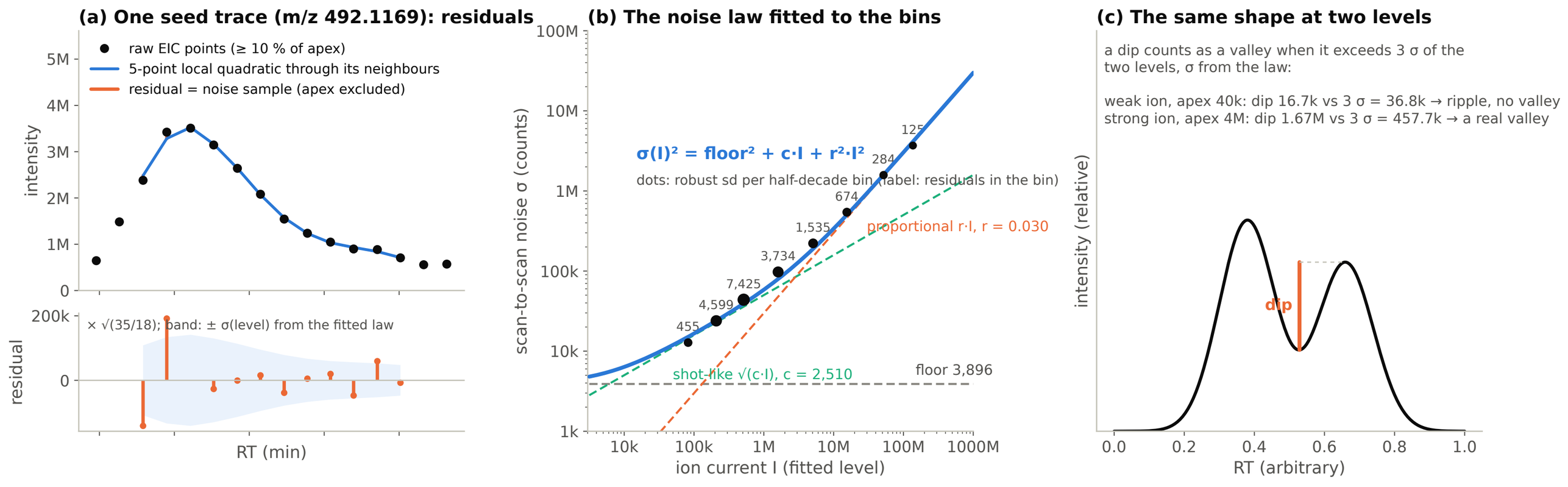
(a) One seed trace (m/z 492.1169): residuals
raw EIC points (≥ 10 % of apex)
5-point local quadratic through its neighbours
residual = noise sample (apex excluded)
intensity
5M
4M
3M
2M
1M
0
residual
200k
0
× √(35/18); band: ± σ(level) from the fitted law
RT (min)
(b) The noise law fitted to the bins
σ(I)² = floor² + c·I + r²·I²
dots: robust sd per half-decade bin (label: residuals in the bin)
scan-to-scan noise σ (counts)
100M
10M
1M
100k
10k
1k
455
4,599
7,425
3,734
1,535
674
284
125
proportional r·I, r = 0.030
shot-like √(c·I), c = 2,510
floor 3,896
10k
100k
1M
10M
100M
1000M
ion current I (fitted level)
(c) The same shape at two levels
a dip counts as a valley when it exceeds 3 σ of the
two levels, σ from the law:
weak ion, apex 40k: dip 16.7k vs 3 σ = 36.8k → ripple, no valley
strong ion, apex 4M: dip 1.67M vs 3 σ = 457.7k → a real valley
intensity (relative)
dip
0.0
0.2
0.4
0.6
0.8
1.0
RT (arbitrary)